

Week 8 Weekly Report

Week 8 integrated the complete project into a self-contained chooser pricing application. The tool exposes the Week 4 BSM model, the selected Week 6 volatility-plus-BSM route, the direct-price benchmark, unit-labelled Greeks, sensitivity analysis, performance evidence, and live-data status.

1 Work Completed

- Bundled the validated BSM engine, trained models, model metadata, test predictions, cached market snapshot, and training-domain ranges.
- Implemented dual pricing with BSM and the selected Random Forest volatility-plus-BSM route.
- Added a 90% proxy-error band calibrated from purge-aware out-of-fold training predictions and evaluated on the test period.
- Added input-domain warnings when live spot, rate, volatility, or VIX falls outside the Week 5 model range.
- Built a four-tab Streamlit dashboard for pricing, sensitivity, performance, and limitations.
- Ran notebook integration checks, an online refresh test, a server health check, and an automated Streamlit page test.

2 Deliverables

- Week8_Final_Integration_and_Validation.ipynb
- self-contained pricing_tool folder
- dual-pricing and stress-test outputs
- deployment checklist
- 5–10 minute demo script
- Final Project Report
- Final Presentation

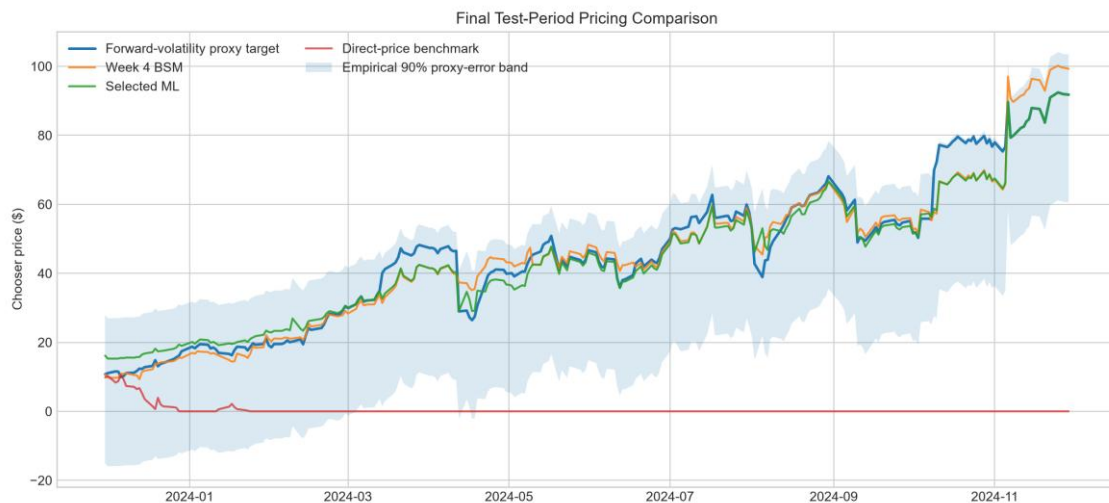
3 Progress Summary

The corrected technical pipeline is complete. All eight final integration checks passed. The nominal 90% proxy-error band covered 99.60% of the 252 later test observations and had mean width \$42.8744. The bundled offline example uses the final Week 2 observation so the submitted artefacts remain reproducible without a network connection; the app can refresh current market inputs when run online.

4 Experimental Results

Route or interval	MAE	RMSE or coverage	R ² or width
Week 4 BSM	3.5698	4.8706	0.9460
Approach 1 RF vol plus BSM	3.1384	4.3203	0.9575
Approach 2 direct price	45.0275	49.9891	-4.6915
Empirical 90% band	—	99.60% test coverage	42.8744 mean width

Figure 1 Final held-out pricing comparison



5 Issues Encountered

- The bundled offline example uses a JPM spot of \$231.08, which exceeds the actual fitted-sample maximum of \$152.49. The application displays an ML extrapolation warning; BSM remains available without this data-driven range restriction.
- A hosted public URL requires the student's GitHub and Streamlit account authorization.
- The required narrated demo video requires the student's recording and submission. A timed script and checklist are included.

6 Plan for Next Stage

- Push the pricing_tool folder to the student's repository and connect it to Streamlit Community Cloud.
- Record the 5–10 minute demonstration using the supplied script.
- Submit the final report, presentation, code, model files, and prototype URL.